AUTOBOTs

AUTONOMOUS SECURITY ENGINEER

EXECUTIVE SUMMARY

AUTOBOTs is an AI-driven autonomous penetration testing platform that performs the complete pen-testing lifecycle with minimal human input. Powered by Large Language Models (LLMs), a multi-agent architecture, it achieves a **98.7%** detection rate, **2.1%** false positive rate, and up to **144×** faster results than manual testing.

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Figure 1: Targeted SDGs

PROBLEM STATEMENT

Manual security testing is cost & time intensive and produces inconsistent & error prone results.

- **High Time & Cost**
- **Manual Effort**
- **Inconsistent Results**
- **Error Prone**

Figure 2: Problems

METHODOLOGY

1. **Input & Target Definition**: User defines scope via UI/API
2. **AI-Driven Reconnaissance**: Automated OSINT & attack surface mapping
3. **Intelligent Analysis:** LLM interprets data & identifies vulnerabilities
4. **Autonomous Execution**: Scanning, exploitation & PoC validation
5. **Real-Time Adaptation**: Dynamic strategy based on findings
6. **Reporting & Remediation**: Verified results with actionable insights

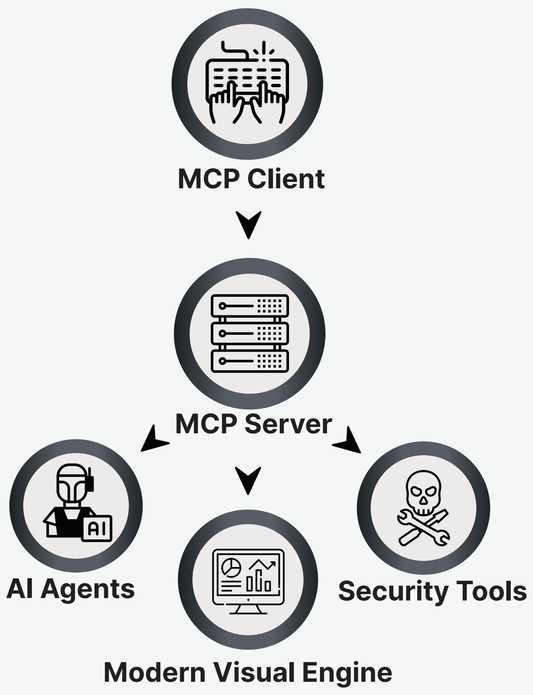

Figure 3: Process Diagram

CONCEPT VALIDATION

AUTOBOTs was evaluated within a controlled test environment consisting of intentionally vulnerable web applications and services to simulate real-world security scenarios. The system successfully demonstrated its ability to autonomously perform end-to-end penetration testing. The results can be seen below:

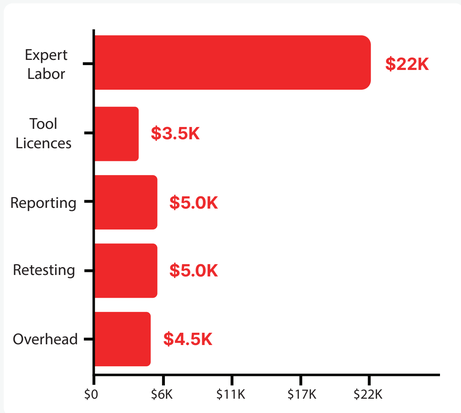

Figure 4: Manual Security Test (Cost)

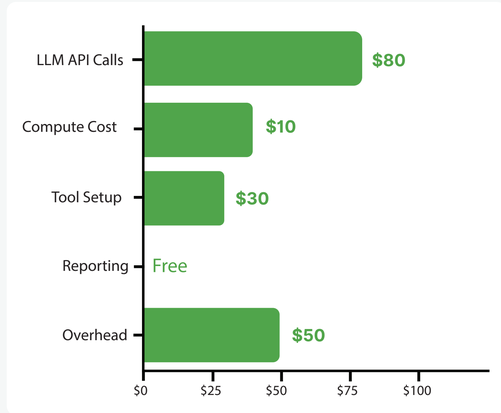

Figure 5: AUTOBOTs Testing (Cost)

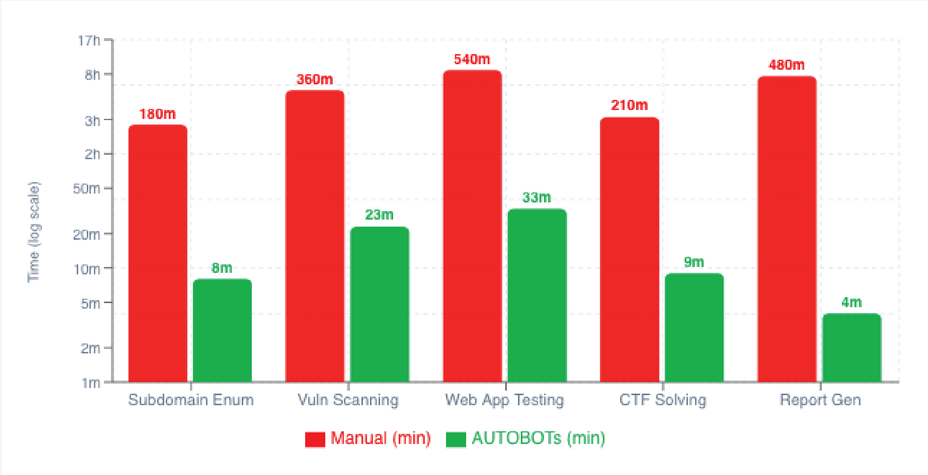

Figure 6: Time Comparison

SOFTWARE & TOOLS

CONCLUSION

AUTOBOTs proves that multi-agent, LLM-powered penetration testing outperforms both manual methods and existing tools in speed, accuracy, and scalability.

ACKNOWLEDGEMENT

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